INCH-POUND

A-A-59131A
19 March 2004
SUPERSEDING
A-A-59131
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COMMERCIAL ITEM DESCRIPTION

DETECTOR, METAL FLAW, ELECTRONIC

The General Services Administration has authorized the use of this Commercial Item Description for all federal agencies

1. Scope. This Commercial Item Description (CID) defines the minimum performance and technical requirements for a lightweight microprocessor-based eddy current flaw detection instrument suitable for Nondestructive Inspection (NDI) use in a military aircraft maintenance environment, which may be at fixed military bases, deployed to austere forward operating locations, or naval vessels afloat.

2. Classification. There are four classifications of metal flaw detectors.

Type I – Kit with bolthole scanner and accessories for Air Force organizations.
Type II – Kit with accessories but without bolthole scanner for Air Force organizations
Type III – Kit with bolthole scanner and accessories for Naval Air Systems Command organizations.
Type IV – Kit with accessories but without bolthole scanner for Naval Air Systems Command organizations.

Comments, suggestions, or questions on this document should be addressed to AFRL/MLS-OL, 4750 Staff Drive, Bldg. 3220 – Door 1, Tinker AFB OK 73145-3317. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at www.dodssp.daps.mil.

AMSC N/A FSC 6635

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3. Salient Characteristics:

3.1 General. The unit shall use a microprocessor-based, impedance plane display, and have dual frequency capability.

3.2 Foreign Object Damage. The unit shall be designed to prevent loss of components, which could generate foreign object damage to aircraft or engines.

3.3 Size and Weight. The unit size shall not exceed 432 cubic inches, with any dimension exceeding 12 inches. The weight shall not exceed 7 pounds with batteries installed.

3.4 Power Requirements. The unit shall operate when supplied with either 100-250 Volt Alternating Current – 50 or 60 Hertz (100-250 VAC – 50/60 Hz) automatically selected and connected directly, through means of a battery charger connected to 100-250VAC-50/60Hz, or batteries/battery pack(s) as described below.

3.4.1 Battery Charger. The unit shall be supplied with a separate battery charger that is able to simultaneously charge the batteries/battery pack(s) and supply power to operate the instrument (if required).

3.4.2 Battery Duration. The cells in the rechargeable battery pack(s) shall not have a charge "memory" limitation that causes the unit to operate for less than the specified time, and provide at least 8 hours of continuous use without scanner or display backlighting, and at least 4 hours of continuous use with both scanner and backlighting at 68 degrees Fahrenheit. The unit must incorporate a visible means of indicating the charge condition of the battery pack(s) and shall provide a low battery warning in order to perform a post-calibration before the battery is exhausted. A minimum of two (2) complete sets of batteries/battery pack(s) shall be provided, so that one set may be charging while the other set is powering the instrument.

3.4.3 Battery Charging. The batteries shall be easily rechargeable without tools and require no longer than 4 hours to charge from 50% to 100% capacity, nor more than 8 hours for a complete charge.

3.4.4 Battery Replacement. The unit shall be capable of continuous 24 hour per day – 7 day per week operation, by swapping rechargeable battery packs by the user without tools and within 90 seconds while meeting the requirements of paragraph 3.3. If designed for use with other than common commercially available worldwide (e.g. AA or D cell) batteries, an auxiliary battery pack designed for AA or D cells shall be provided, which shall connect to the flaw detector. The auxiliary battery pack is not subject to the requirement of paragraph 3.3. If designed for proprietary battery packs, the government shall acquire the design specifications and a limited license for manufacture after contract expiration.

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3.5 Controls. The unit function controls shall be menu driven. As a minimum, the unit shall accommodate control of the following functions.

3.5.1 Frequency. The operating frequency shall be selectable from 60 Hz to 6 MHz (minimum range), with a precision of two significant digits (i.e. 1 Hz steps from 60-100Hz, 10 Hz steps from 100Hz-1kHz, 100Hz steps from 1-10kHz, etc). The frequency synthesizer shall be accurate to within five (5) percent of the selected value. The unit shall be capable of dual frequency operation.

3.5.2 Gain. The unit shall be capable of at least 85 dB of amplification range for both the vertical (y) and horizontal (x) axis.

3.5.3 Probe Drive. The unit probe drive circuit shall allow the equipment operator to vary the signal driving the probe independent of gain and frequency adjustment.

3.5.4 Phase Rotation. The unit shall be able to rotate the eddy current impedance vector, as viewed on the display, through a full 360 degrees in one degree or smaller increments in either the clockwise or counter-clockwise directions.

3.5.5 Filters. The unit shall be equipped with high pass, low pass, and bandpass filters. The low pass filter shall be selectable from 10 Hz to 500 Hz minimum range, and the high pass filter shall be selectable from 0 to 200 Hz minimum range.

3.5.6 Alarm Function. The unit shall incorporate an alarm (gate) system that will trigger an audible and/or visual alarm when the test signal enters or leaves a designated alarm zone. The alarm function shall be adjustable for size, position, and shape, and useable in both the impedance plane and sweep modes.

3.5.7 Programmable Test Procedures. The unit shall have the capability to program, store, and recall a minimum of 50 instrument test parameter settings. The instrument memory shall enable alphanumeric labeling of programs. Programmed instrument settings shall be adjustable after recall. The memory shall retain the settings for a minimum of 300 hours after electrical power is turned off and primary batteries are removed.

3.5.8 Screen Capture. The unit shall have the capability to freeze and store at least 20 screen displays for subsequent review or download to PC or printer.

3.6 Instrument Display. The display shall have a minimum area of 13 square inches with a minimum resolution of 76,800 total pixels. The display shall be easily seen both in bright light (i.e. flightline ~ 500 foot-candles) or low light (i.e. hanger ~ 10 foot-candles) conditions. The display shall have a selectable graticule for both major and minor divisions.

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3.7 Persistence. The display persistence shall be variable and adjustable from 0.1 to 10 seconds minimum range, or until cleared by the operator.

3.8 Display Modes. The unit shall display eddy current inspections signals in the impedance plane (flying dot), the vertical amplitude versus time (sweep), and dynamic scanner modes.

3.8.1 Impedance Plane Mode. The impedance plane mode shall display the eddy current signal as an impedance vector representing the vector sum of the test coil inductive reactance and resistance. The impedance plane mode shall be available for both manual and bolthole scanner modes. The display shall incorporate separate controls for horizontal and vertical positioning of the balance point anywhere on the screen.

3.8.2 Sweep Mode. The sweep mode shall display the vertical impedance plane component of the eddy current signal as a function of time. The unit shall be capable of synchronizing with an external signal.

3.8.3 Dynamic Mode. The display shall have a dynamic mode that is compatible with the scanner defined in paragraph 3.10.3. The dynamic display shall automatically initiate a sweep trace at the start of each scanner probe rotation and use a waterfall type display to show consecutive sweeps or show only the current sweep while automatically erasing the previous sweep information. The full 360 degrees of rotation shall be displayed regardless of any nominal changes in rotation speed of the scanner probe.

3.9 Conductivity and Coating Thickness Measurement. The unit shall be capable of measuring electrical conductivity and non-conductive coating thickness as specified below. The unit shall come equipped with appropriate probes and cables for conductivity and thickness measurement. The probe(s) shall be replaceable in the field without the need for depot or contractor intervention.

3.9.1 Electrical Conductivity. The unit shall be capable of measuring electrical conductivity of materials in the range from 1 to 65% IACS (International Annealed Copper Standard). The unit shall be capable of using 60 kHz for conductivity measurement. The measurement accuracy for the range from 20 to 65% IACS shall be within $\pm$ 0.5% IACS at 68 degrees Fahrenheit with liftoff up to 0.008 inches. The accuracy for the range from 1 to 65% IACS shall be within $\pm$ 1.0% IACS from 32 to 100 degrees Fahrenheit.

3.9.2 Non-conductive Coating Thickness. The unit shall be capable of measuring non-conductive coatings up to 0.025 inches thick. The measurement accuracy shall be within $\pm$ 5% of the reading.

3.10 External Connections. The unit shall accommodate the following external connections.

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3.10.1 Probe Interface. The selection of a probe interface that uses eddy current probes currently in the government inventory without an adapter, or can be easily adapted with minimal sensitivity losses.

3.10.2 Probe Compatibility. The unit shall be electronically compatible with all currently available commercial off-the-shelf absolute, reflection, and differential-type probes.

3.10.3 Scanner Interface. The unit shall interface and be compatible with bolthole scanners. As a minimum, the eddy current unit shall provide the following: (1) scanner power, (2) sweep coordination (external triggering), (3) control of scanner probe revolutions per minute (RPM), (4) compatibility with remote null switching on the scanner, and (5) capability to change probes in the field. The scanner control capability shall be included with the unit and be ready for use. Types II and IV shall include a high-speed bolthole scanner in the kit.

3.10.4 Analog Output. The unit shall have analog outputs to allow for horizontal and vertical component test signal monitoring.

3.10.5 Computer Interface. The unit shall include Microsoft Windows 2000 (or latest release) compatible software and cables to enable communication with a personal computer. Software capabilities shall include (1) software version updates, (2) uploading and downloading instrument program settings, and (3) data reporting utilities such as downloading screen captures.

3.10.6 Printer Output. The unit shall be capable of printing screen presentations and test procedures. This may be accomplished either directly to a printer or through a Microsoft Windows compatible computer, with supplied software. The unit shall come with drivers for common commercially available printers, including the Hewlet-Packard Laserjet series.

3.11 Accessories.

3.11.1 Protective Carry Cases. The unit shall include soft-shell and hard-shell carrying cases. The soft shell case shall protect the unit from damage during use while still allowing access to all external connections. The hard-shell case shall be suitable for storing and shipping the unit with accessories. If the hard-shell case with unit and accessories weighs in excess of 25 pounds, the hard shell case shall include rollers and collapsible handle typically found on luggage.

3.11.2 Cables. All types shall have a cable to connect absolute “Microdot” type probes to the unit. Types III and IV shall include a dual BNC adapter cable.

3.11.3 Probes. The unit shall have a conductivity/coating thickness probe to meet the requirements of paragraph 3.9. Probe diameter shall not exceed one-half (0.5) inch.

3.11.4 Auxiliary Battery Pack. An auxiliary battery pack shall be provided which utilizes either “AA” or “D” cell batteries if not used as the primary battery.

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3.12 Environmental Conditions. The unit shall be operable in the following environmental conditions.

3.12.1 Explosive Atmospheres. The unit shall be tested and certified by a reputable laboratory for safe use, while operating on batteries, in an explosive atmosphere as defined by Class 1, Division 2, Group D, in the National Fire Protection Association code (NFPA 70E), Section 500. The testing method shall be in accordance with MIL-STD-810, Method 511.4, Procedure I.

3.12.2 Operating Conditions. The unit and accessories shall perform to the specification in this CID throughout temperatures from 15 to 120 degrees Fahrenheit at 95% relative humidity and from sea level to 7280 feet above sea level.

3.12.3 Storage Conditions. The unit and accessories shall perform to the specifications in this CID following a seven (7) day exposure to temperatures from 0 to 140 degrees Fahrenheit at 95% relative humidity.

3.13 Calibration. The unit shall be user calibrated without the need for manufacturer's periodic calibration. No factory recalibration decals shall be applied to the unit.

3.14 Software Revisions. Software revisions shall be user installable from a Microsoft Windows 2000 (or latest release) based personal computer. All software revisions made during the life of the contract shall be included at no additional charge.

3.15 Documentation. One copy of the commercial operation and maintenance instructions, including calibration procedures and adjustments, as applicable, shall be shipped with each item. Two paper and two electronic copies of the operation and maintenance instructions (above), illustrated parts breakdown, diagnostics, and servicing instructions, shall be sent to the Contracting Officer for delivery to the Equipment Specialist and cognizant Engineer. The documentation shall be releasable for third-party repairs after expiration of the contract.

3.16 Warranty. A one-year warranty, to include parts, labor, and two-way shipping, shall be included against manufacturing defects.

3.17 Options. A listing of available accessory items (including bolthole scanners), extended warranties, and standard repairs shall be provided to the Contracting Officer.

4. Regulatory Requirements. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

4.1 Metric Products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pounds unit, provided they fall within specified tolerances using conversion tables contained in the latest revision of Federal Standard No. 376,

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and all other requirements of this Commercial Item Description are met. If a product is manufactured to metric dimensions and those dimensions exceed tolerances specified in the inch-pound units, a request should be made to the contracting officer to determine if the product is acceptable. The contracting officer has the option of accepting or rejecting the product.

5. PRODUCT CONFORMANCE PROVISIONS.

5.1 Product Conformance. The products provided shall meet the salient characteristics of this Commercial Item Description, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial marketplace. The government reserves the right to require proof of such conformance.

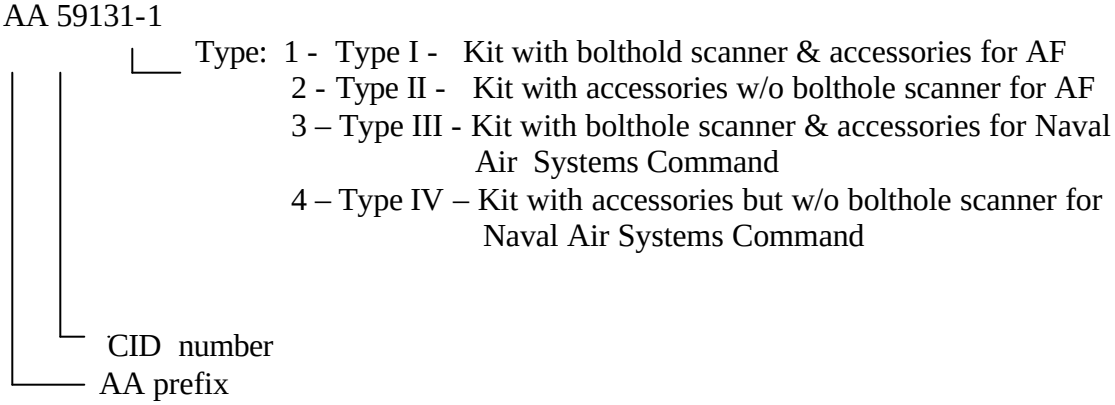
5.2 Market Acceptability. All of the following market acceptance criteria are necessary to document the quality, reliability, and supportability of the product to be provided under this specification. (1) The company producing the item must have been producing eddy current flaw detectors for at least the past three years. (2) The company must have sold the basic commercial unit in the industrial market for a minimum of one year. (3) The company must have sold at least fifty (50) units similar to the units being offered, with no major defects within the last three (3) years. A major defect is a defect, which affects the performance of the unit and requires an engineering change to the unit's original configuration in order for the unit to perform the requirements of this CID.

6. PACKAGING

6.1 Preservation, Packaging, Packing, Labeling, and Marking. Preservation, packaging, packing, labeling, and marking shall be specified in the contract or order.

7. Notes.

7.2 Part Identification Number (PIN). The following part identification numbering procedure is for government purposes and does not constitute a requirement for the contractor.



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7.3 Sources of documents

7.3.1. Government Documents. Military Specifications, Standards, and Handbooks reference herein may be obtained from the Standardization Documents Order Desk, 700 Robbins Ave., Bldg 4, Section D, Philadelphia, PA 19111-5094.

7.4 Ordering data.

7.4.1 Title, number, and date of this description.

7.4.2 Specify special packaging requirements.

7.4.3 Contracting officer may request proof of certification of commercial item prior to first contract delivery.

7.5 Key Words.

bolthole scanner
dual frequency
eddy current
electrical conductivity
electromagnetic testing
impedance plane

7.6 National Stock Number.

NSN
6635-00-463-1574RN

7.7 Value Added Features. As described in the synopsis/solicitation, the government may pursue contracting by negotiation leading to selection of the proposal representing the best value to the Government per FAR part 15. Evaluation factors may include (but not limited to) features such as additional test modes, reduction in size & weight, extended battery duration, battery charging enhancements (reduced charge time, conditioning, multiple batteries), expanded frequency range or sweep, expanded signal filtering, expanded alarm functions, expanded instrument programming/data storage, expanded display options, expanded electrical conductivity frequency and range, industry standard probe interface(s), expanded scanner compatibility, expanded computer/printer interface capability, additional soft/hard case protection, enhanced auxiliary battery pack, Electro-Magnetic Interference (EMI) protection, expanded operating/storage temperature range, self-calibration/self-test capability, extended warranty, and firm-fixed pricing of accessories.

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MILITARY INTEREST:

Preparing Activity:
Air Force- 51

Custodian:
Air Force - 99
Navy - AS
DLA - GS
Army - GL

Agent:
Air Force - 99

Project 6635-0207

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